

Calibrated Scenario Report

Purpose

The first sensitivity analysis showed that the default Mesa simulation was strongly biased toward escalation. That was useful because it revealed how the current model behaves, but it did not give enough variety for scientific explanation.

This second calibration experiment was designed to find different behavioural regimes:

- calming or very low escalation
- low escalation
- medium escalation or unresolved risk
- high escalation
- tipping-point scenarios where outcomes are mixed

The goal is to help explain how different combinations of bystander roles and harmful-content pressure affect the simulated environment over time.

What Changed From the First Report

The first report mostly used the dashboard's standard settings:

- starting bullying level around 45
- toxicity around 50
- moderate profanity and identity attack
- balanced or moderately varied bystander mixes

Those settings created strong upward pressure, so most scenarios escalated.

In this calibration experiment, I tested lower and middle harm levels more carefully. This lets the model show a wider range of outcomes.

Mesa Simulation Method

The same Mesa backend was used:

- mesa_bridge.py
- mesa_model.py
- mesa_agents.py
- mesa_learning.py

Each scenario was repeated across multiple random seeds. This means the results are averages across repeated simulation runs, not one single example.

The core stopping rules stayed the same:

Condition	Outcome
bullying level reaches 80 or above	Got worse
bullying level reaches 20 or below	Calmed down
maximum steps reached without either threshold	Stayed unresolved

Default Model Settings Kept Constant

Unless a scenario changed them, these settings were kept the same:

Parameter	Value
total bystanders	24
maximum steps	18
ToM influence strength	0.55
learning rate	0.30
reward strength	0.90
memory retention strength	0.80
adaptation speed	0.35
carry learning across independent runs	False
escalation threshold	80
calming threshold	20

Broad Calibration Grid

The first calibration grid tested:

```text

6 harm profiles x 5 bystander profiles x 30 repeated runs

```

This produced:

```text

900 Mesa simulation runs

```

Harm Profiles

Harm profile	Starting bullying	Toxicity	Profanity	Identity attack	Likes	Retweets
Very low harm	15	0	0	0	5	5
Low harm	25	10	5	5	10	10
Mild harm	35	15	10	10	20	20
Medium harm	40	25	15	20	25	25
Upper-medium harm	45	35	25	30	35	35
High harm	55	50	35	45	50	50

Bystander Mix Profiles

Bystander profile	Bully supporters	Victim supporters	Silent bystanders	Other
Protective majority	5%	70%	20%	5%
Defender leaning	15%	55%	25%	5%
Balanced discussion	25%	35%	30%	10%
Silent majority	15%	20%	55%	10%
Bully-support leaning	45%	20%	25%	10%

Broad Calibration Findings

The broad grid showed clear high and very-low regimes, but the low and medium regimes needed a finer search.

Very Low / Calming Examples

Scenario	Mean final level	Change	Got worse	Calmed down	Unresolved	Mean steps
Very low harm + Protective majority	14.4	-0.6	0%	100%	0%	1.0
Very low harm + Defender leaning	14.7	-0.3	0%	100%	0%	1.0
Very low harm + Balanced discussion	15.6	+0.6	0%	100%	0%	1.0
Very low harm + Silent majority	15.8	+0.8	0%	100%	0%	1.0
Very low harm + Bully-support leaning	16.3	+1.3	0%	100%	0%	1.0

Medium / Unresolved Examples

Scenario	Mean final level	Change	Got worse	Calmed down	Unresolved	Mean steps
Mild harm + Protective majority	59.9	+24.9	43%	20%	37%	16.0
Low harm + Defender leaning	50.0	+25.0	17%	30%	53%	16.2

High Escalation Examples

Scenario	Mean final level	Change	Got worse	Calmed down	Unresolved	Mean steps
High harm + Protective majority	84.3	+29.3	100%	0%	0%	4.2
High harm + Balanced discussion	85.0	+30.0	100%	0%	0%	3.5
High harm + Silent majority	85.2	+30.2	100%	0%	0%	3.7
Medium harm + Bully-support leaning	84.2	+44.2	100%	0%	0%	6.8

Fine Calibration Search

The broad grid showed that low and medium escalation happen around a tipping point. To find those cases more clearly, I ran a finer search.

This search tested:

```text

588 parameter combinations x 30 repeated runs

```

This produced:

```text

17,640 Mesa simulation runs

```

The fine search varied:

- starting bullying level from 22 to 42
- toxicity from 0 to 30
- defenders from 50% to 80%
- instigators from 5% to 20%

- neutral bystanders as the remaining percentage

Recommended Calibrated Scenarios

These are the best scenarios to use in your project presentation because they show different outcomes clearly.

Scenario A: Low Escalation / Mostly Controlled

Parameter Settings

Parameter	Value
starting bullying level	38
toxicity	5
profanity	2.5
identity attack	3
likes	5
retweets	5
bully supporters	15%
victim supporters	60%
silent bystanders	20%
other	5%

Results

Metric	Value
mean final bullying level	34.6
mean change	-3.4
got worse	0%
calmed down	40%
stayed unresolved	60%

Interpretation

This is a low-escalation scenario. The bullying does not usually become severe. The strong defender presence prevents the situation from getting worse, but it does not always fully calm the conversation.

In simple terms:

People defending the victim keep the situation under control, but the conversation may still remain tense or unresolved.

Scenario B: Low Escalation With Balanced Pressure

Parameter Settings

Parameter	Value
starting bullying level	38
toxicity	0
profanity	0
identity attack	0
likes	5
retweets	5
bully supporters	20%

Parameter	Value
victim supporters	50%
silent bystanders	25%
other	5%

Results

Metric	Value
mean final bullying level	35.1
mean change	-2.9
got worse	0%
calmed down	50%
stayed unresolved	50%

Interpretation

This scenario shows low escalation when the harmful-content level is very low. Even with some bully supporters, the lack of toxicity makes the situation easier to contain.

In simple terms:

When the original situation is not very harmful, defenders can keep the conversation from becoming dangerous.

Scenario C: Medium Escalation / Unresolved Risk

Parameter Settings

Parameter	Value
starting bullying level	42
toxicity	10
profanity	5
identity attack	6
likes	10
retweets	10
bully supporters	20%
victim supporters	60%
silent bystanders	15%
other	5%

Results

Metric	Value
mean final bullying level	54.6
mean change	+12.6
got worse	47%
calmed down	3%
stayed unresolved	50%

Interpretation

This is a medium-risk scenario. Even though defenders are the majority, the situation often remains unresolved and sometimes escalates.

In simple terms:

Defenders help, but the starting bullying level and harmful content are high enough that the situation is still risky.

Scenario D: Medium Mixed Outcome

Parameter Settings

Parameter	Value
starting bullying level	38
toxicity	10
profanity	5
identity attack	6
likes	10
retweets	10
bully supporters	15%
victim supporters	50%
silent bystanders	30%
other	5%

Results

Metric	Value
mean final bullying level	53.7
mean change	+15.7
got worse	33%
calmed down	13%
stayed unresolved	53%

Interpretation

This scenario is useful because it shows mixed outcomes. The situation does not always escalate, but it often remains unresolved.

In simple terms:

The conversation is balanced on the edge. Some runs get worse, some calm down, and many stay unsettled.

Scenario E: Tipping-Point Scenario

Parameter Settings

Parameter	Value
starting bullying level	38
toxicity	15
profanity	7.5
identity attack	9
likes	15
retweets	15
bully supporters	5%
victim supporters	70%
silent bystanders	20%

Parameter	Value
other	5%

Results

Metric	Value
mean final bullying level	48.8
mean change	+10.8
got worse	33%
calmed down	37%
stayed unresolved	30%

Interpretation

This is the best example of a tipping-point scenario. The same settings can lead to different outcomes because agent activation and action choices vary across runs.

In simple terms:

This is a fragile situation. Strong defenders help, but the harmful content is strong enough that the conversation can still go either way.

Scenario F: High Escalation

Parameter Settings

Parameter	Value
starting bullying level	55
toxicity	50
profanity	35
identity attack	45
likes	50
retweets	50
bully supporters	25%
victim supporters	35%
silent bystanders	30%
other	10%

Results

Metric	Value
mean final bullying level	85.0
mean change	+30.0
got worse	100%
calmed down	0%
stayed unresolved	0%

Interpretation

This is a high-escalation scenario. Harmful-content pressure is high enough that even a reasonable number of defenders cannot stop escalation.

In simple terms:

When the original harm is severe and engagement pressure is high, the conversation gets worse very quickly.

What This Means Scientifically

The calibrated scenarios show that the model can represent different levels of escalation.

The earlier escalation-heavy result happened because the default settings placed the environment in a high-pressure region of the model.

The new calibration shows a more complete picture:

Calibration type	What it shows
Very low harm	The situation calms quickly because there is little harmful pressure
Low escalation	Defenders control the situation, but it may remain unresolved
Medium escalation	Defenders help, but the situation is still risky
Tipping point	Small differences in agent behaviour can change the final outcome
High escalation	Toxicity and engagement overwhelm bystander defence

Recommended Dashboard Presets

For the dashboard, the most useful presets would be:

Preset name	Purpose
Low-risk controlled conversation	Shows defenders keeping bullying low
Medium-risk unresolved conversation	Shows tension continuing without a clear outcome
Tipping-point conversation	Shows why bystander actions matter
High-risk escalating conversation	Shows harmful-content pressure overwhelming the environment

Honest Limitation

This calibration does not prove that real online conversations behave exactly this way. It shows how the current Mesa model behaves under different assumptions.

That is still valuable because agent-based modelling is about testing assumptions. The calibrated scenarios make the simulation more useful for explanation because they show more than one possible outcome.

Output Files

The calibration experiment produced:

- calibration_run_level_results.csv
- calibration_summary.csv
- calibration_trajectories.csv
- fine_calibration_search.csv
- recommended_calibrations.csv
- CALIBRATED_SCENARIO_REPORT.md